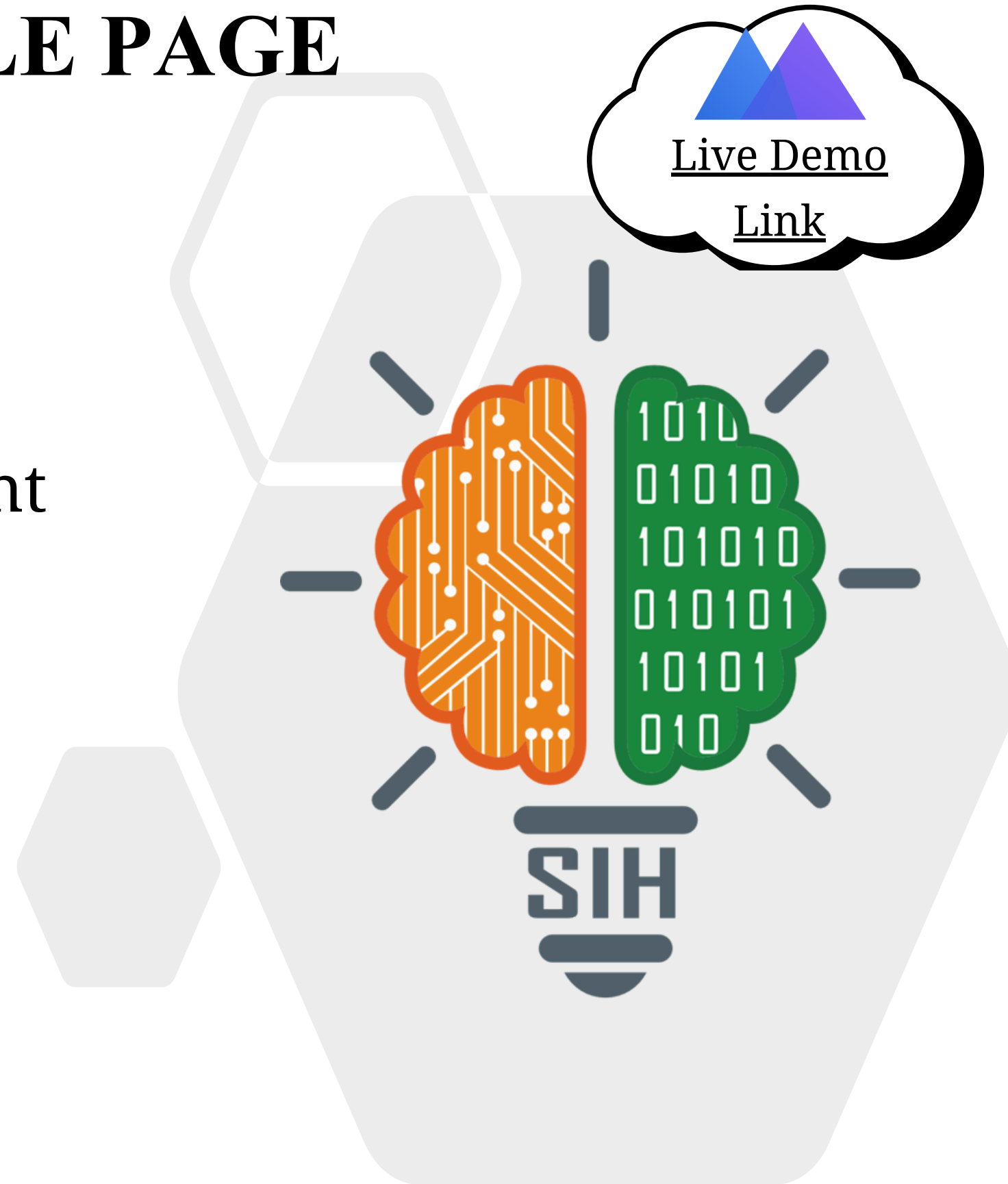


TITLE PAGE

- **Problem Statement ID** – 26175
- **Problem Statement Title** –
DepthWizard Single-View Height
Estimation and 3D Fly-through
- **Theme** – Disaster Management
- **PS Category** – Software
- **Team ID** – 121971
- **Team Name** – Team Endra



Proposed Solution!



Our Solution



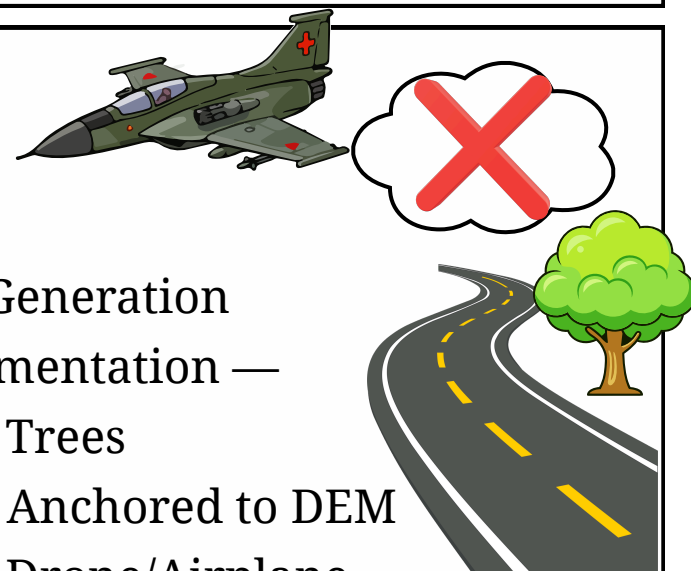
DepthWizard turns a single satellite image into a digital surface model(DSM), anchored to Low resolution DEM for georeferenced scenes, that you can fly through, measure & validate in the browser.



Our Features

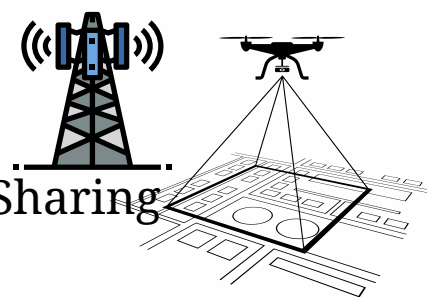
Core Features

1. Immersive DSM Generation
2. Built-in Class Segmentation — Buildings, Roads, Trees
3. Flood Simulation Anchored to DEM
4. 3D Flythrough — Drone/Airplane
5. First-Person Walkthrough
6. Works without internet.

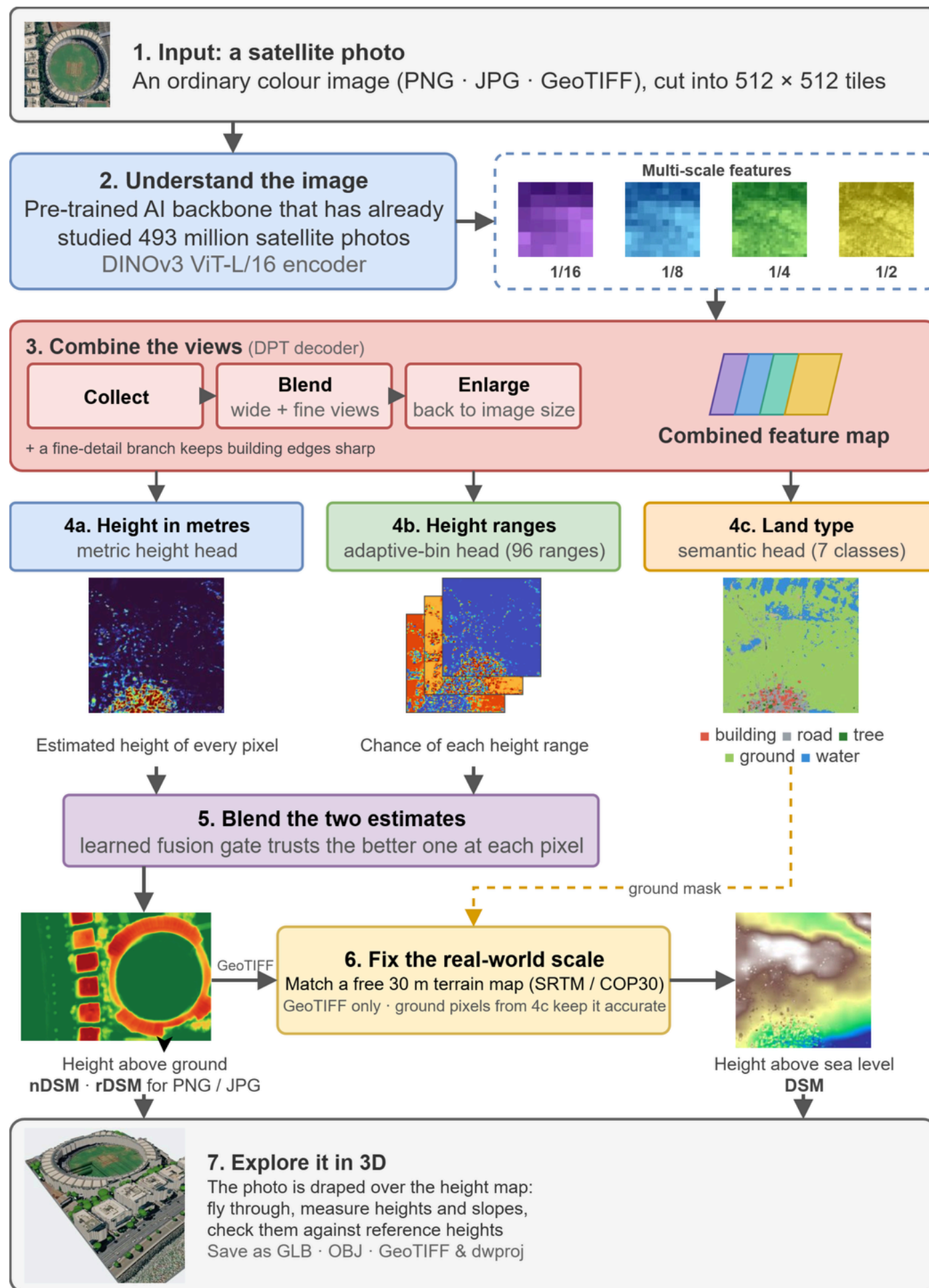


Unique Features

1. Cloud Detection & Masking
2. OpenStreetMap Integration
3. Telecom Tower Coverage Simulation
4. Custom .dwproject Project Sharing
5. Colour-Blind Support



IDEA TITLE



1.1 Model Architecture Flow

Why This?

What is the Problem?

1. **Expensive 3D Mapping:** LiDAR and stereo methods are accurate but require specialized data, sensors, and processing
2. **Need for a Fast, Low-Cost Alternative:** We need elevation estimation directly from readily available optical imagery.

Why Current Solutions Struggle?

1. **General Models Face a Domain Gap:** Most depth models are not specifically designed for top-down satellite imagery. [\[DA v2 VS DINO v3\]](#)
2. **Depth is Often Scale-Ambiguous:** Relative depth does not directly give accurate height
3. **Height Estimation Needs Remote-Sensing Data:** Generic depth models need adaptation for satellite-specific height prediction.

How Our Solution is better?

1. **Satellite-Native Backbone:** We use DINOv3 SAT-493M, pretrained specifically on satellite imagery.
2. **Fine-Tuned for Height:** We train it on remote-sensing datasets to predict above-ground height, not generic depth.
3. **Built for Remote-Sensing Generalization:** The system is optimized for urban, rural, hilly, and vegetated scenes.

Dataset	Why ?	Size	Link
GAMUS	Urban aerial RGB + heights; benchmark	64GB	Kaggle Source
SynRS3D	Synthetic, exact heights; urban + terrain	23GB	Kaggle Source
DFC2019 (Track1)	WorldView-3 + LiDAR; urban, sparse, trees	20.4GB	Kaggle Source
DFC2023 (Track 2)	Global (17 cities across 6 Continents)& Indian (New Delhi) topography contex	12GB	Kaggle Source
MVS3DM (IARPA)	Cartosat like 4-band + LiDAR; validation	22.6GB	Kaggle Source
NEON AOP	LiDAR canopy; forested + hilly landscapes	~16GB	Kaggle Source

- Elevation Extraction**
PyTorch, DINOv3 ViT-L/16, SAT-493M (satellite-pretrained) backbone ONNX Runtime on a Hugging Face Space
- Scale Calibration to DSM**
SRTM / Copernicus 30 m DEM or GCPs
rDSM for PNG/JPG
Backend : FastAPI, Rasterio
- 3D Flythrough & Validation**
React + TypeScript + Three.js , Docker

Tech Stack

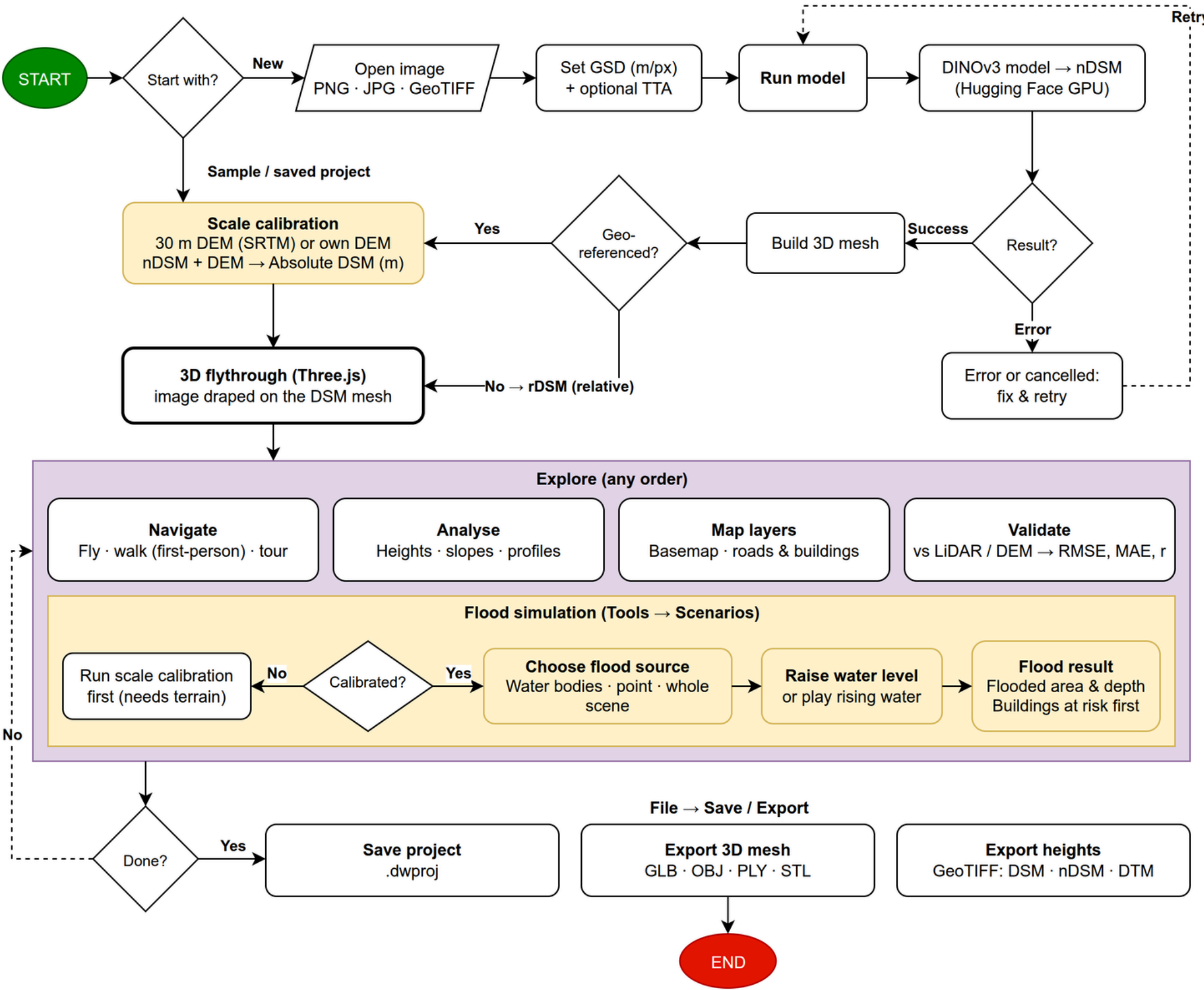


Figure 3 :User Usage Flowchart

Potential Challenges & our Solutions

Tall-Structure Underestimation

Model under-predicts tall buildings



Solution:

1. Height-balanced loss + soft-binned head for heights up to 150 m
2. Tall-building bias tracked every epoch
3. Shadow-length scale check using sun angle

Limited Indian Data & Global Context

Indian imagery with per-pixel heights; LiDAR is scarce



Solution:

1. Satellite-native [DINOv3 ViT-L/16, SAT-493M](#) trained on 493M global Sat images
2. Used the [DFC23](#) dataset covering 17 cities on 6 continents accross the world (Incl New Delhi)

Good Buildings, Weak Forests

Edges over-smoothed trees are the worst class (RMSE 4.19 m)



Solution:

1. Full-resolution branch + NEON forest LiDAR
2. Removed incorrect 0 m tree labels from DFC 2023



Clouds Hide the Ground

Clouds were mistaken for tall objects

Solution:

1. Automatic cloud detection
2. Clouds are automatically detected, masked, and excluded from inference.
3. Marked on the 3D view and left out of inference.



DSM Accuracy vs LiDAR / Reference, by Landscape

[\[Metrics\]](#) [\[Source\]](#)

Landscape	RMSE (m) ↓	MAE (m) ↓	Pearson r ↑	GAMUS RMSE (m)
Urban	3.85	1.69	0.90	3.06
Sparse	1.32	0.32	0.63	1.25
Forested	2.85	1.60	0.84	2.05
Overall	3.04	1.32	0.89	2.59

**Pooled over 4 held-out sets (~184 M pixels): MVS3DM, NEON, GAMUS, US3D.*



Regulatory & Legal Feasibility

1. **Open-Source Stack:** Permissive licences (BSD / MIT / Apache), no fees.
2. **Commercial Use:** DINOv3 licence permits commercial use; training data is CC BY 4.0 / MIT / public.
3. **Satellite Data Policy:** Cartosat data is licensed from NRSC Bhoonidhi under the Indian Space Policy 2023; DEMs are free (SRTM public domain, Copernicus open).
4. **Data Sovereignty:** runs fully offline (ONNX + local DEM)

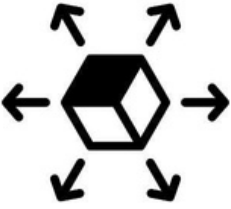


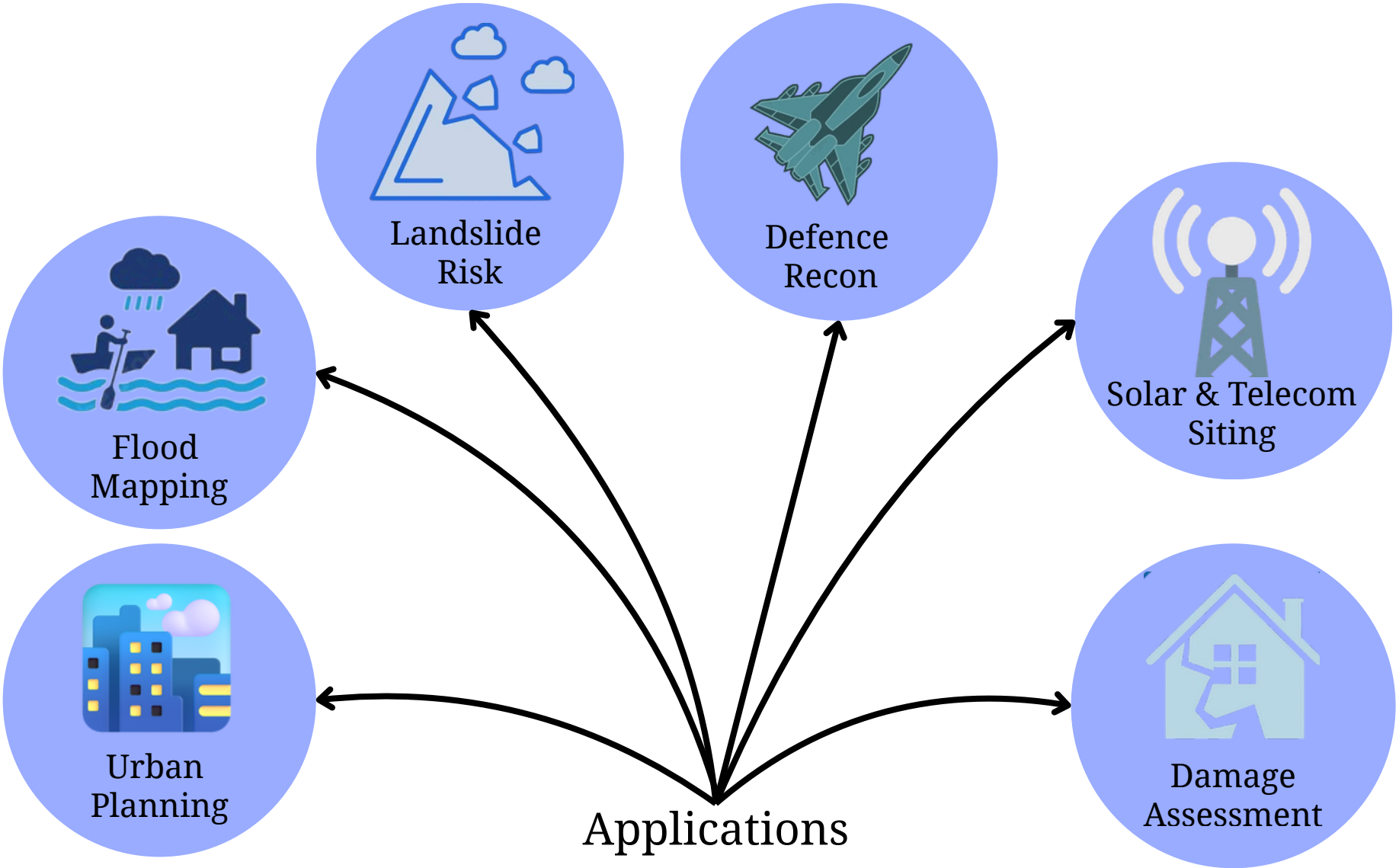
Economic Feasibility

1. **Low Data Cost:** a [Cartosat scene](#) is $\leq$ [₹5,000](#) for ~80 km² (**Cheaper for ISRO**) ; UAV LiDAR for the same area is [~₹5crore](#) +.
2. **Free Elevation References:** SRTM and Copernicus DEMs cost nothing.
3. **Zero Licence Fees:** open-source models and libraries throughout.
4. **Low Running Cost:** CPU inference via ONNX, or a GPU on demand from \$0.59/h ([NVIDIA T4](#)).

IMPACT AND BENEFITS

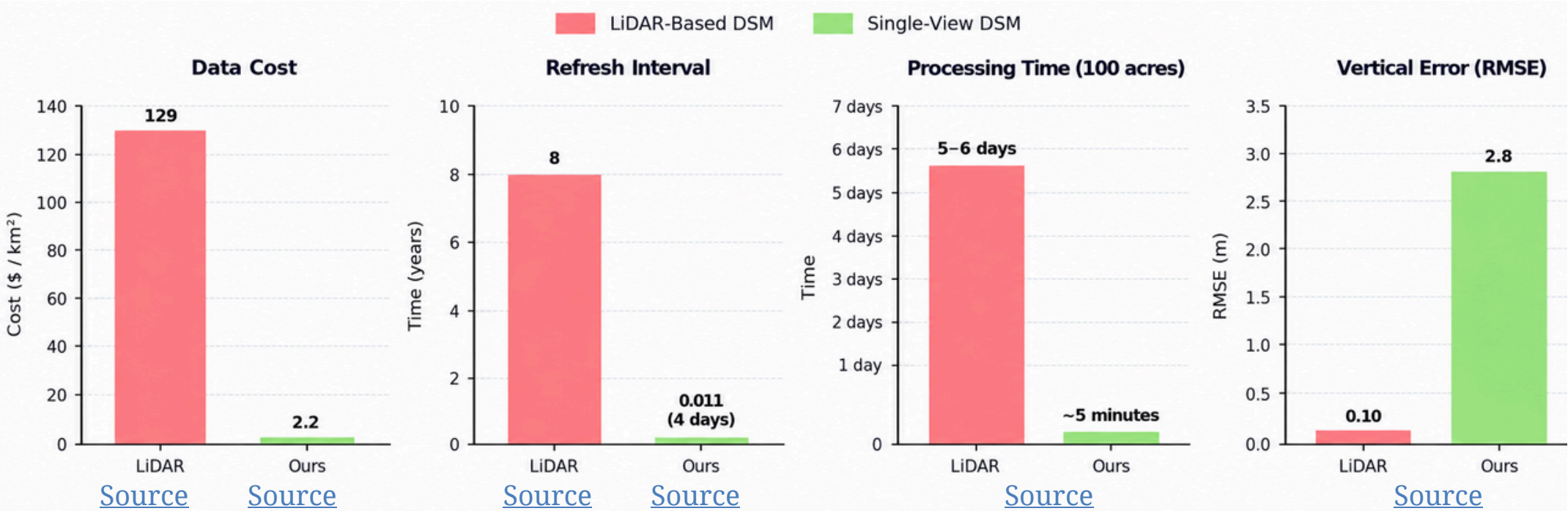
Impacts of Solution

 Faster Disaster Assessment 3D surface from one post-event image, no survey flight needed	 Frequent Updates Refresh maps every satellite pass (~4 days), not every few years
 Nationwide Scale Same pipeline for any district: full 20k × 20k px scenes	 Remote DSM Generation Reaches floods, hills and borders where crews and aircraft can't
 Decision-Ready Insight Officials measure heights, slopes and profiles in the browser	 Unlocks ISRO Archive Turns Cartosat imagery since 2016 into 3D history for before/after studies



Benefits of Solution

 Cost Effective Imagery ~59× cheaper per km² than airborne LiDAR	 Faster & Scalable GPU inference in tiles; web app, ONNX export for offline use
 3D Flythrough Orbit, walk and drone views with slope and profile tools	 Wide Coverage Any area with satellite imagery; ~81 km² per Cartosat scene
 Built-in Validation Upload any reference DSM → RMSE, MAE and r on screen	 Metric Heights GeoTIFF → absolute DSM anchored to SRTM / Copernicus DEM



Model

DINOv3Self-supervised
Vision
Foundation
Model[source](#)**Bhuvan**Platform to Access
your data and
create governance
applications[source](#)

Platform

Model

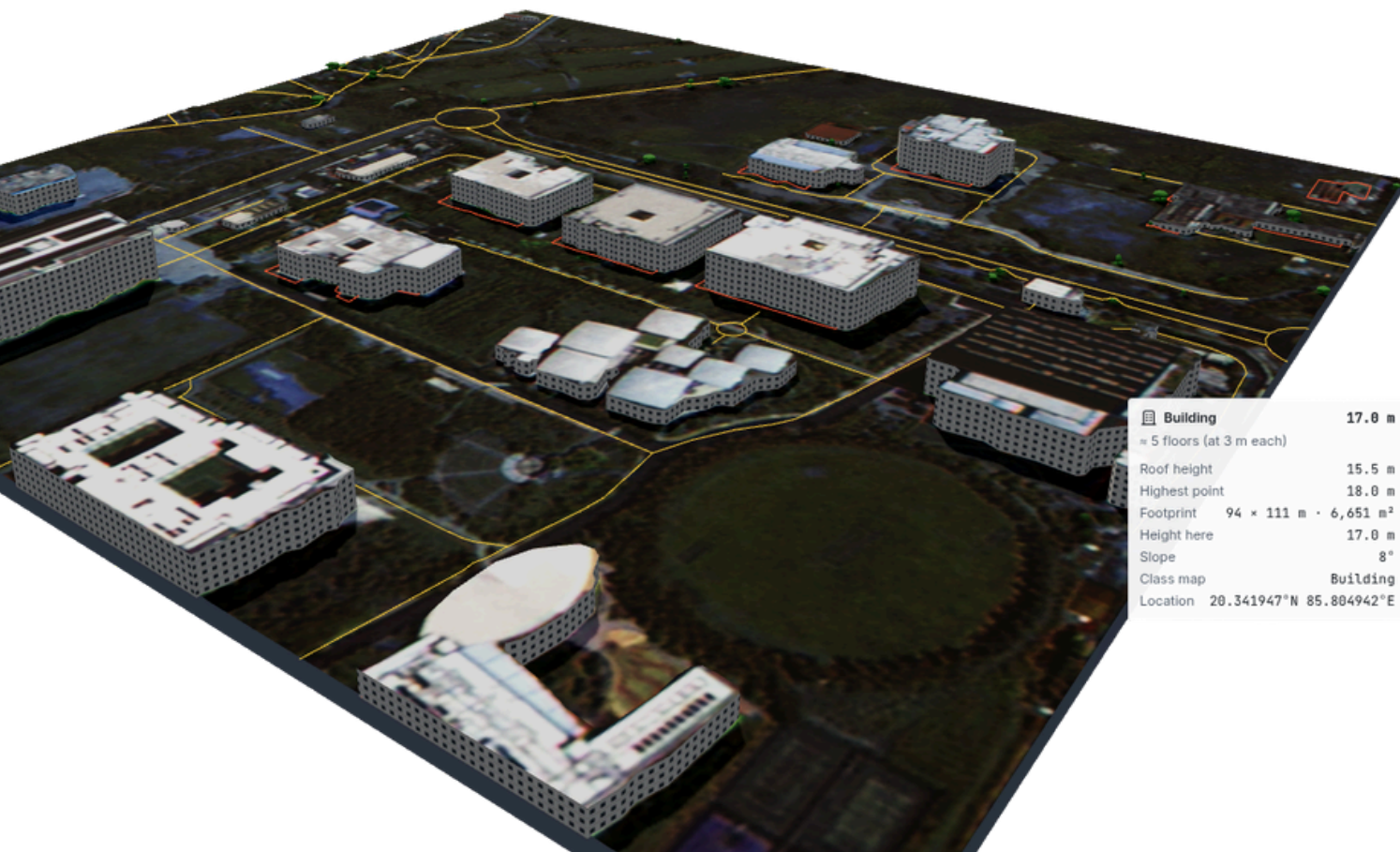
Monocular
depth estimation
(MDE)**Depth Anything
V2**[source](#)Access to our
extensive archive
of Remote Sensing**Bhoonidhi**[source](#)

Platform

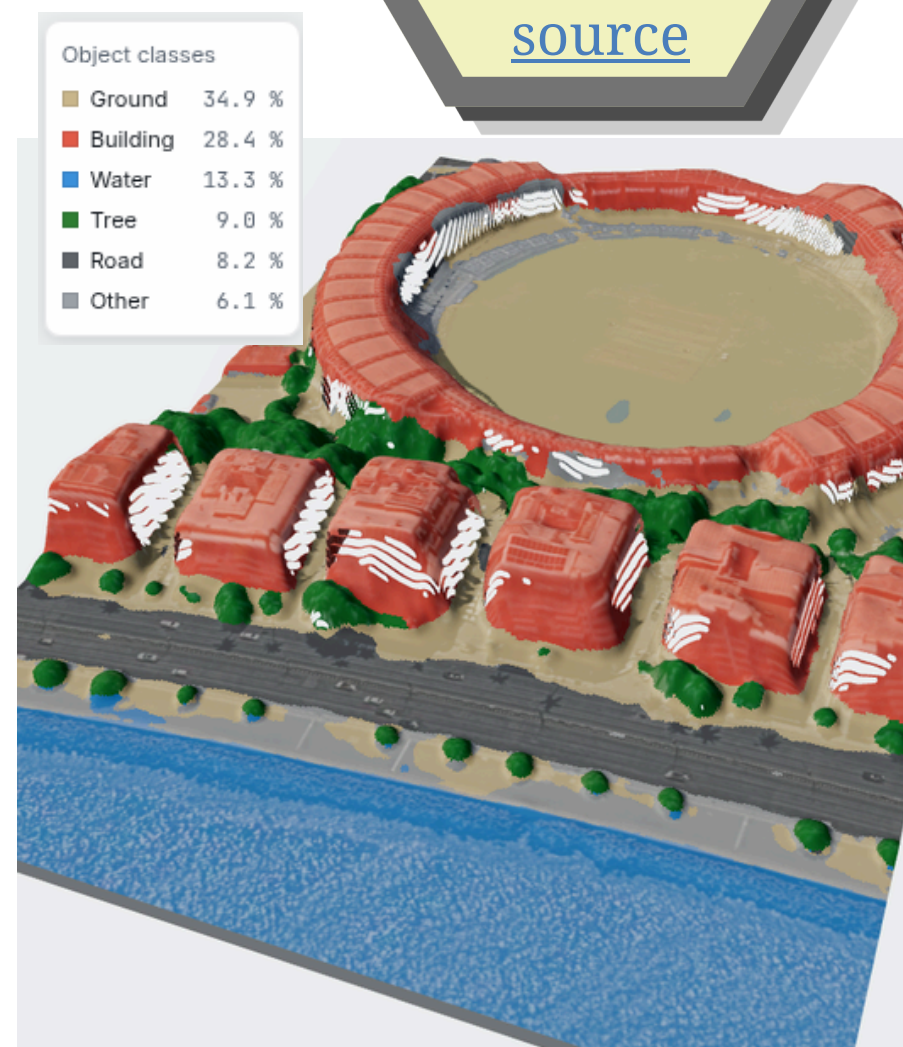
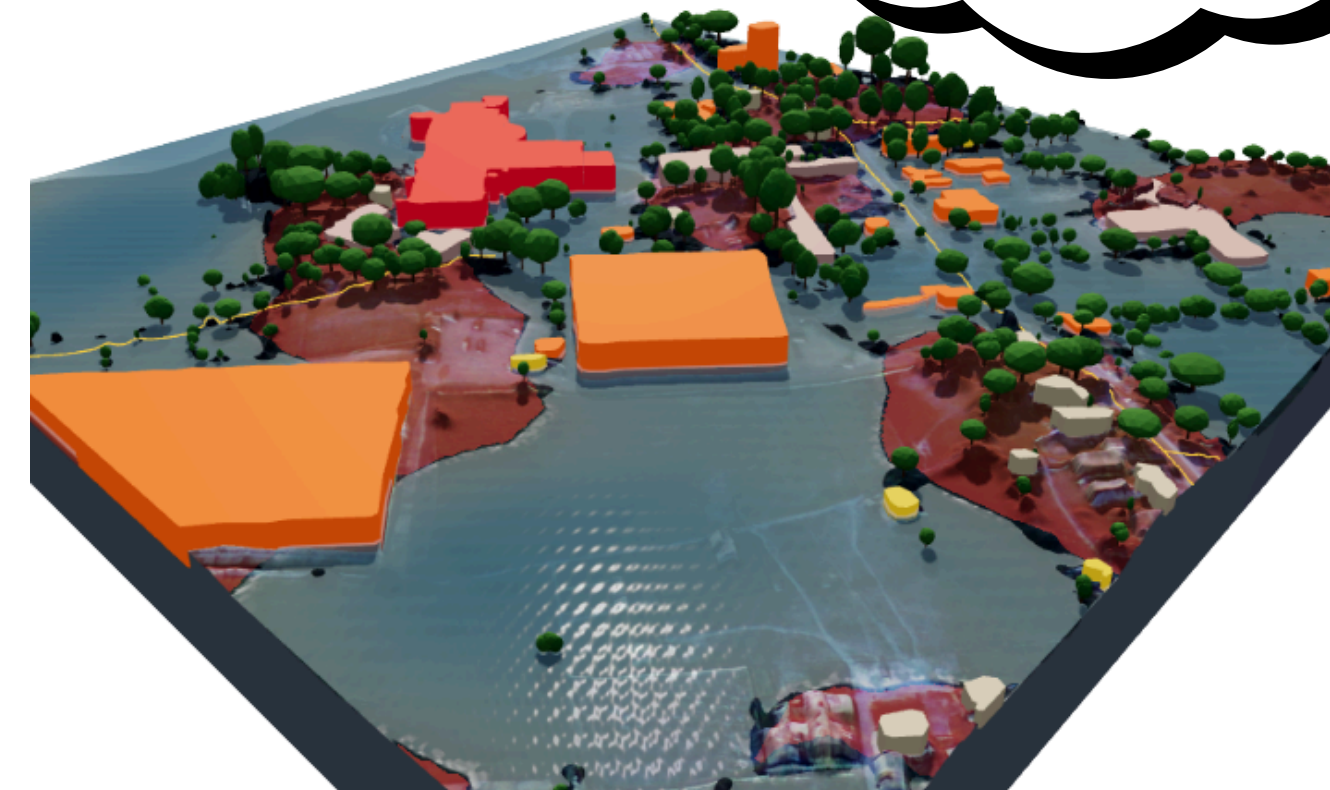
Research
paper**CHMv2**Improvements in
Global Canopy
Height Mapping
using DINOv3[source](#)**PyTorch**PyTorch is an
optimized tensor
library for deep
learning[source](#)

Library

Library

RasterioPython library for
processing
geospatial raster
data like GeoTIFFs
and DEMs.[source](#)**Other
References**For all
references visit
our docs website[source](#)1. [Github](#)2. [Technical
Report](#)**Georeferenced & Post-Processing**

Infosys, Bhubaneswar .tif from Cartosat-2S

[\[Google Maps\]](#) [\[Source\]](#)**Ungeoreferenced & Built in Segmentation**DSM Model of Wankhede Stadium Mumbai From a
Screen-Shot [Location](#)**Georeferenced**Cartosat 3 : Flood Simulation at [Bani, Uttar Pradesh](#)
[Location](#) [\[Source\]](#)